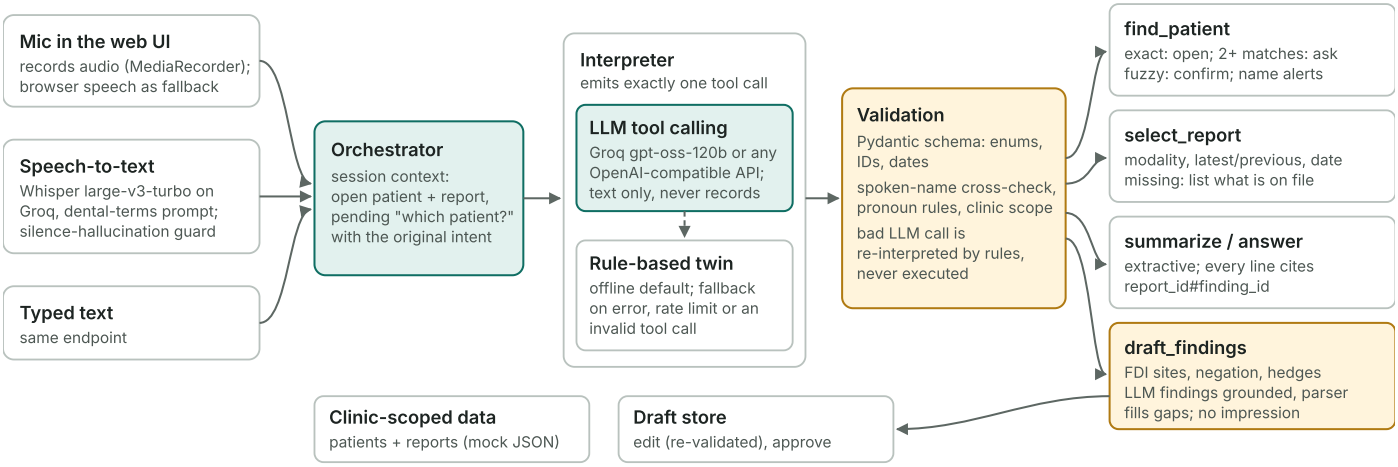
Viscan Voice Dental Assistant: architecture

Proof of concept · Python / FastAPI · synthetic data only · LLM interprets, code decides



FLOW

1. **Voice to text.** The UI mic records audio and posts it to `/api/assistant/voice`, which transcribes it with Whisper on Groq and rejects Whisper's silence hallucinations ("Thank you."). Browser speech recognition is the fallback; tests use a mock. From here on, voice and typed input are handled identically.
2. **LLM interpretation.** The LLM (Groq `openai/gpt-oss-120b`, or any OpenAI-compatible API by config) receives the utterance and three static tool schemas (`report_request` with open/summarize/question, `draft_findings`, `unsupported_request`) with tool choice forcing one call. It is told only whether a patient is open, never names, IDs or report text. No key, a timeout, a rate limit or an unusable call falls back to the rule-based interpreter, which emits the same internal calls.
3. **Business-rule validation.** Arguments are schema-checked; a bad LLM call is never executed, and a spoken name the model dropped is restored. Deterministic tools then decide identity (two Priyas: ask; "Pria": confirm; exact match: open with a look-alike alert), context ("her" only if a female patient is open), clinic scope, and report selection.
4. **Report generation.** Dictation becomes findings with FDI site, qualifiers, present/absent, stated/uncertain, the exact evidence quote and review reasons. LLM findings must pass grounding: quote is in the transcript, quote mentions the finding, tooth and negation match the words, qualifiers were spoken. Rejections are shown with reasons; parser findings the model missed are merged in; unmapped speech is kept verbatim; unspecified sides stay unresolved. No impression is generated, and approval needs a linked patient and resolved teeth.

PRODUCTION CONSIDERATIONS

Concern	This PoC	Next step
Privacy	LLM sees no records; audit log stores IDs only; free tiers used with synthetic data only	In-region / self-hosted STT and LLM, zero-retention terms, encrypted drafts, PHI-scrubbed logs (DPDP Act)
Authorization	Clinic-scoped reads; out-of-scope patients look non-existent; sessions bound to a clinician; drafts private	OIDC tokens, role-based sign-off, break-glass access logging
Latency	Measured live: STT 350–480 ms, interpreter about 1 s, tools about 1 ms; LLM capped at 8 s with instant rule fallback	Streaming STT; rules-first routing so the LLM handles dictation and misses only; cache tool definitions
Cost	Groq free tier: $0 within rate limits. At Groq's paid rates, about $0.0003 per spoken command (interpreter plus STT)	Prompt caching; self-hosted Whisper and open-weight model at volume
Failures	LLM error or rate limit → rule fallback; STT → clear 4xx; size limits; no stack traces to clients	Retry budgets, circuit breaker, alerts on rejection and clarification rates

Code: app/interpreter (LLM + rules), app/validation.py (grounding), app/tools (retrieval, dictation, drafts), app/orchestrator.py (context). 57 automated tests plus 4 live API tests. Prices from Groq's published rates, mid-2026.